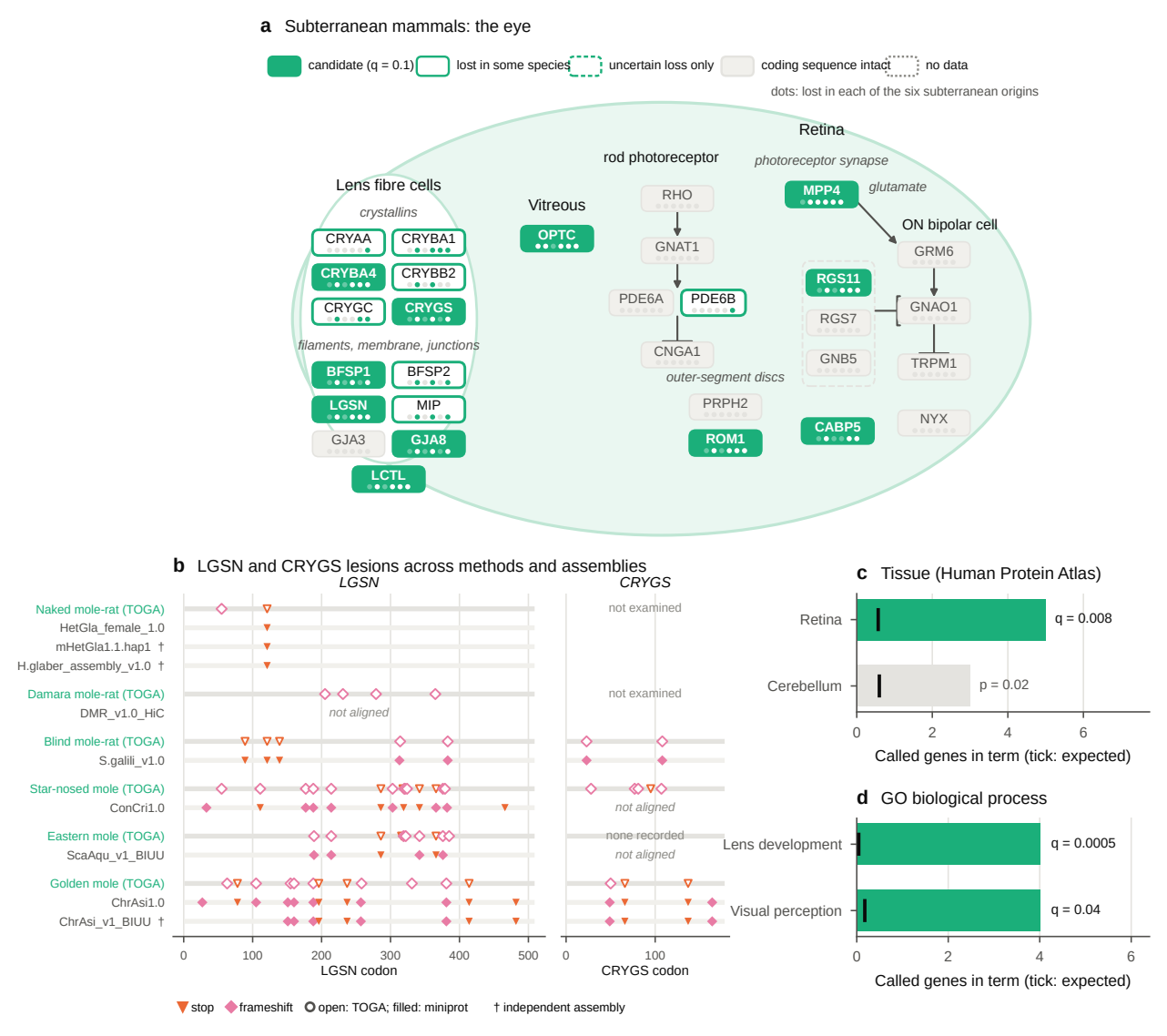

a Subterranean mammals: the eye
candidate (q = 0.1)
lost in some species
uncertain loss only
coding sequence intact
no data
dots: lost in each of the six subterranean origins
Retina
rod photoreceptor
photoreceptor synapse
Lens fibre cells
crystallins
Vitreous
RHO
MPP4
glutamate
ON bipolar cell
CRYAA
CRYBA1
OPTC
GNAT1
GRM6
CRYBA4
CRYBB2
RGS11
PDE6A
PDE6B
GNAO1
CRYGC
CRYGS
RGS7
filaments, membrane, junctions
CNGA1
GNB5
TRPM1
BFSP1
BFSP2
outer-segment discs
LGSN
MIP
PRPH2
NYX
GJA3
GJA8
ROM1
CABP5
LCTL
b LGSN and CRYGS lesions across methods and assemblies
LGSN
CRYGS
Naked mole-rat (TOGA)
HetGla_female_1.0
mHetGla1.1.hap1 †
H.glaber_assembly_v1.0 †
Damara mole-rat (TOGA)
DMR_v1.0_HiC
not aligned
not examined
Blind mole-rat (TOGA)
S.galili_v1.0
Star-nosed mole (TOGA)
ConCri1.0
Eastern mole (TOGA)
ScaAqu_v1_BIUU
none recorded
Golden mole (TOGA)
ChrAsi1.0
ChrAsi_v1_BIUU †
0
100
200
300
400
500
LGSN codon
CRYGS codon
stop
frameshift
open: TOGA; filled: miniprot
† independent assembly
c Tissue (Human Protein Atlas)
Retina
Cerebellum
q = 0.008
p = 0.02
Called genes in term (tick: expected)
d GO biological process
Lens development
Visual perception
q = 0.0005
q = 0.04